

CAIRNGORM SUMMIT, United Kingdom

WMO: 030650

Lat: 57.1163N

Lon: 3.6439W

Elev: 4085

StdP: 12.65

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	13.8	15.5	-10.7	3.6	32.1	0.0	6.4	33.5	84.3	27.9	80.2	29.6	28.3	40	1.557

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.7	59.4	50.4	56.4	49.2	53.7	49.0	53.7	55.8	52.0	54.0	50.4	52.6	19.7	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
53.0	69.5	54.1	51.2	65.0	52.4	49.4	60.7	50.9	24.0	55.7	22.9	54.0	21.9	52.8	66.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
92.1	76.4	67.1	12.2	64.1	2.9	3.9	10.2	66.9	8.5	69.2	6.9	71.4	4.8	74.2
			11.5	58.3	2.9	4.3	9.4	61.5	7.7	64.0	6.0	66.4	3.9	69.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	34.4	26.5	26.4	27.7	32.2	36.2	40.0	44.8	43.7	40.9	35.6	30.6	27.6	(6)
(7)		DBStd	8.91	5.26	6.77	6.46	6.50	7.66	6.17	5.66	5.06	5.51	5.47	5.54	6.15	(7)
(8)		HDD50	5753	728	662	693	533	437	308	183	207	279	447	581	695	(8)
(9)		HDD65	11168	1193	1082	1158	983	894	749	625	659	723	911	1031	1160	(9)
(10)		CDD50	59	0	0	0	0	8	9	23	13	6	0	0	0	(10)
(11)		CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
(12)		CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)
(13)		CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	32.8	36.9	34.5	33.9	31.8	26.1	25.9	26.1	28.7	31.9	38.1	38.5	40.7	(14)
(15)	Precipitation	PrecAvg	35.3	3.7	2.6	2.5	2.2	2.4	2.6	2.7	3.1	2.8	3.6	3.5	3.7	(15)
(16)		PrecMax	46.3	8.9	7.2	6.3	5.1	4.4	5.5	5.7	6.7	9.0	5.8	6.3	8.8	(16)
(17)		PrecMin	24.6	0.7	0.5	0.4	0.4	0.8	0.9	0.9	1.1	0.9	0.9	0.9	0.8	(17)
(18)		PrecStd	4.5	1.9	1.5	1.3	1.1	0.9	1.1	1.2	1.3	1.5	1.4	1.1	1.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.0	48.7	49.5	52.3	61.3	62.7	62.3	62.2	57.9	52.4	47.9	45.9	(19)
(20)			MCWB	38.1	41.7	40.9	46.4	51.6	49.8	52.6	54.5	46.7	41.7	42.6	39.8	(20)
(21)		2%	DB	40.1	43.9	43.7	47.3	56.6	57.2	58.8	56.6	54.1	48.3	43.9	42.1	(21)
(22)			MCWB	37.9	38.2	36.5	41.3	48.6	47.1	50.3	52.2	46.8	44.0	39.3	39.0	(22)
(23)		5%	DB	37.3	39.6	39.3	44.1	51.3	52.2	55.9	53.5	51.6	45.9	41.2	39.5	(23)
(24)			MCWB	36.5	37.4	36.3	39.5	47.2	47.2	50.1	50.3	47.6	43.0	39.2	37.3	(24)
(25)		10%	DB	34.5	36.2	36.6	41.3	47.2	48.8	53.4	51.0	49.0	43.7	38.6	36.8	(25)
(26)			MCWB	33.4	35.0	34.9	38.2	43.4	45.0	49.8	48.7	46.6	41.1	37.1	35.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.9	46.4	43.0	49.6	54.0	54.1	55.8	57.4	52.9	48.6	44.7	43.8	(27)
(28)			MCD B	43.0	46.8	45.1	49.6	57.2	58.1	57.2	60.5	53.9	49.2	45.3	44.2	(28)
(29)		2%	WB	38.4	41.1	39.1	44.0	50.4	51.1	53.4	53.5	50.7	46.3	41.3	40.4	(29)
(30)			MCD B	38.6	42.1	40.7	45.3	54.5	54.2	55.4	55.3	51.7	46.6	41.6	40.9	(30)
(31)		5%	WB	36.0	38.0	37.0	41.1	47.5	48.1	51.9	51.3	48.6	44.3	39.0	38.0	(31)
(32)			MCD B	36.4	39.1	38.1	42.7	51.2	50.7	54.1	52.8	49.9	44.9	39.4	38.4	(32)
(33)		10%	WB	33.4	35.3	34.7	38.5	44.0	45.7	50.0	49.2	46.7	42.1	36.8	35.1	(33)
(34)			MCD B	34.0	36.2	35.9	40.1	46.6	48.2	52.7	50.5	48.2	42.8	37.2	35.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.9	6.9	7.4	7.9	8.4	7.7	7.7	7.0	7.1	7.0	6.9	7.2	(35)
(36)			MCDBR	12.1	11.4	11.7	11.5	11.2	11.4	10.3	10.1	9.8	11.0	11.3	12.5	(36)
(37)			MCWBR	10.9	10.6	10.1	9.1	9.3	9.9	8.3	8.7	8.9	10.0	10.6	11.3	(37)
(38)		5% WB	MCD BR	11.4	11.3	11.4	10.2	10.9	10.1	9.5	9.6	9.2	10.6	10.4	12.2	(38)
(39)			MCWBR	11.1	11.3	10.6	9.3	9.2	9.4	8.2	8.8	9.3	10.5	10.3	11.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.250	0.271	0.291	0.306	0.309	0.309	0.311	0.301	0.290	0.279	0.254	0.232	(40)	
(41)		taud	2.336	2.461	2.462	2.467	2.492	2.508	2.500	2.536	2.576	2.561	2.417	2.243	(41)	
(42)		Ebn at Noon	211	258	280	291	295	296	293	290	279	252	210	180	(42)	
(43)		Edh at Noon	14	20	25	30	31	31	30	27	23	18	13	11	(43)	
(44)	All-Sky Solar Radiation	RadAvg	141	345	670	1099	1475	1449	1400	1106	776	412	185	100	(44)	
(45)		RadStd	20	34	77	108	88	125	121	74	62	41	17	12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (37 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page